1. **Identify whether a mathematical result or conclusion does, or does not, make sense given the context**
   Relevance: ★★★★★ (highest)

   The core question for each judgment item is:
   "Can this conclusion necessarily be derived from the given information?"

   It is **not** "Is it possibly true?" but rather **"Is it necessarily true?"**
   Emphasizes the evaluation of **conclusion validity and certainty**.

   👉 This is the **core competency required for the entire question.**

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2. **Understand the relationship between the context of the problem and the representation of the mathematical solution**
   Relevance: ★★★★★

   Examples:

   - "0.01 zeds per kWh" → Does this imply that maintenance costs depend on electricity generation?
   - "97% operational" → Can this be directly converted into "97 days of downtime"?

   Students must understand:

   - What the mathematical representation **actually means**
   - What it **does not mean** in the real-world context

   👉 Prevents **superficial, formulaic interpretations**.

3. **Identify the limits of the model used to solve a problem**
   Relevance: ★★★★☆

   The problem frequently tests:

   - Whether the given information is sufficient
   - Whether there are implicit but unstated conditions

   For example:

   - Do we know the annual electricity generation?
   - Do we know the specific distribution of downtime across the year?

   👉 Emphasizes **"do not exceed what the model and data allow."**

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4. **Recognize the extent and limits of mathematical concepts and solutions**
   Relevance: ★★★★☆

   Even if the calculation is mathematically valid,
   without necessary information, a conclusion **cannot be drawn**.

   Example:

   - Can "approximately 5%" be calculated? Missing annual generation → conclusion cannot be inferred

   👉 Mathematical reasoning ≠ arbitrary inference.

5. **Interpret a mathematical result back into the real-world context**
   Relevance: ★★★☆☆

   Requires translating:

   - Unit costs
   - Percentages
   - Annual operational ratios

   into real-world meaning,
   which is **particularly critical when judging "whether a conclusion can be drawn."**

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6. **Construct and communicate explanations and arguments in the context of the problem**
   Relevance: ★★★☆☆

   Students usually need to explain mentally or in writing:

   - Why "Yes"
   - Why "No"

   Emphasizes **logical justification rather than numerical computation**.

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7. **Use knowledge of how the real world impacts outcomes to make contextual judgments**
   Relevance: ★★★☆☆

   Examples:

   - 97% operational efficiency ≠ 97 fixed days of downtime
   - Downtime may be intermittent
     Real-world operation affects whether a conclusion can be inferred

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8. **Interpret mathematical outcomes in a variety of formats or compare representations**
   Relevance: ★★☆☆☆

   Involves:

   - Price vs. cost
   - Percentage vs. absolute days

   Does **not** involve charts or multiple complex representations.

9. **Use mathematical tools or computer simulations to test reasonableness**
   Relevance: ★☆☆☆☆ (lowest)

   This question is entirely about:

   - Logical judgment
   - Information interpretation

   No computational tools or simulations are required.

① Identify whether a mathematical result or conclusion does, or does not, make sense given the context

（判断结论在给定情境中是否成立）

相关性：★★★★★（最高）

- 每一个判断项的核心问题都是：

  > “这个结论是否**一定**能从给定信息中推出？”

- 不是“是否可能为真”，而是“是否**必然成立**”
- 强调对结论**合理性与确定性**的判断

👉 这是整道题的**核心能力要求。**

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② Understand the relationship between the context of the problem and representation of the mathematical solution

（理解情境与数学表述之间的关系）

相关性：★★★★★

- 例如：
  - “0.01 zeds per kWh” → 是否意味着维护费与发电量有关？
  - “97% operational” → 是否可以直接换算成“97 天不运转”？
- 学生必须理解：
  - 数学表述**真正意味着什么**
  - 它在现实语境中**不意味着什么**

👉 防止“表面代换式理解”。

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③ Identify the limits of the model used to solve a problem

（识别模型与数据的边界）

相关性：★★★★☆

- 题目多次考查：
  - 给定信息**是否足够**
  - 是否存在**隐含但未说明的条件**
- 如：
  - 是否知道年发电量？
  - 是否知道全年具体停机分布？

👉 强调**“不能超出模型和数据说话”**。

④ Recognise the extent and limits of mathematical concepts and solutions

（认识数学结论的适用范围）

相关性：★★★★☆

- 即使计算在数学上成立
- 但若缺少必要信息，就**不能下结论**
- 例如：
  - “约 5%”是否能算？缺少年发电量 → 结论不可推出

👉 数学推理 ≠ 任意推断。

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⑤ Interpret a mathematical result back into the real world context

（将数学信息解释回现实）

相关性：★★★☆☆

- 需要把：
  - 单位成本
  - 百分比
  - 年度运行比例

    解释为现实含义
- 尤其在判断“是否可以推出”时非常关键

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⑥ Construct and communicate explanations and arguments in the context of the problem

（在情境中进行解释与论证）

相关性：★★★☆☆

- 学生通常需要在脑中或书面上说明：
  - 为什么“Yes”
  - 为什么“No”
- 强调逻辑说明，而非数值计算

⑦ Use knowledge of how the real world impacts outcomes to make contextual judgments

（利用现实知识进行情境判断）

相关性：★★★☆☆

- 例如：
  - 97% 的运行效率 ≠ 固定 97 天停机
  - 停机时间可能是零散的
- 现实运行方式影响结论可否推出

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⑧ Interpret mathematical outcomes in a variety of formats or compare representations

（比较或理解不同表述）

相关性：★★☆☆☆

- 涉及：
  - 价格 vs 成本
  - 百分比 vs 绝对天数
- 但不涉及图表或多重复杂表征

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⑨ Use mathematical tools or computer simulations to test reasonableness

（使用技术工具检验合理性）

相关性：★☆☆☆☆**（最低）**

- 本题完全是：
  - 逻辑判断
  - 信息理解
- 不需要计算工具或模拟